Trees, Binary Search Trees and Heaps

LumoraAI Presentation Demo • 01 Data Structures Algorithms

Binary Search Trees

• A binary search tree maintains the ordering property that values in the left subtree are smaller than the node key and values in the right subtree are larger, subject to the duplicate policy.

• Search follows one root-to-leaf path. Its time depends on tree height: O(h). A balanced BST therefore supports O(log n) search, while a skewed tree can degrade toward O(n).

Heaps

• A binary heap is a complete binary tree represented efficiently in an array. In a min-heap, every parent is less than or equal to its children.

• The root gives constant-time access to the minimum, while insertion and root removal take O(log n) because the element may move along a height-bounded path.

Useful comparison

• BSTs are useful when ordered traversal and search are required. Heaps are useful when the primary operation is repeatedly obtaining the minimum or maximum. Choosing between them should follow the required operations rather than the data structure name alone.

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